Inorganic CUME

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Graphene (a single graphite-sheet, also called "carbon in two-dimension" or "thinnest material possible") is a very hot candidate in today's materials chemistry, physics, and engineering. The questions below might help you organize your thoughts if you were asked to join a meeting on discussing future projects involving graphene.

1. What should be the basic chemical and physical properties for a large sheet of graphene? Why so? (20')

2. Theoretically, what would happen to the graphene's structure and properties if some of its 6-member rings would be replaced by 5-member rings and 8-member rings? (20')

3. Predict, using you chemistry reasoning, why (or why not) the graphene would be most likely toxic (or biocompatible). (20')

4. Propose graphene-based new materials that may help you develop new solar-cell, new rechargeable battery, and new photocatalyst. (30')